

PROJECT REPORT

Heart Disease Analysis

1. INTRODUCTION

1.1 Project Overview

Heart Disease Analysis is a data analytics project that focuses on examining patient health records to identify patterns, trends, and key risk factors associated with heart disease. The project uses data preprocessing, exploratory data analysis, and interactive Tableau dashboards to provide meaningful healthcare insights for better decision-making.

1.2 Purpose

The purpose of this project is to analyze heart disease data, identify important health indicators, and present the findings through interactive visualizations. The dashboard helps healthcare professionals understand patient conditions and supports data-driven decisions.

2. IDEATION PHASE

2.1 Problem Statement

Heart disease is one of the leading causes of death worldwide. Healthcare organizations generate large amounts of patient data, but extracting meaningful insights manually is difficult. There is a need for an interactive analytical dashboard that helps identify disease patterns and risk factors efficiently.

2.2 Empathy Map Canvas

Users

- Doctors
- Healthcare Professionals
- Hospital Management
- Medical Researchers

Needs

- Quick access to patient insights
- Easy-to-understand visual reports

- Identification of high-risk patients
- Better healthcare planning

Problems

- Manual analysis is time-consuming
- Difficulty in identifying hidden trends
- Large volume of patient records

Expectations

- Interactive dashboards
- Accurate visual analysis
- Easy decision-making support

2.3 Brainstorming

Possible solutions discussed:

- Collect patient health data
- Clean and preprocess the dataset
- Perform exploratory data analysis
- Create interactive Tableau dashboards
- Generate healthcare insights and reports

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Collect patient health records.
2. Clean and prepare the dataset.
3. Analyze important health parameters.
4. Create Tableau visualizations.
5. Generate insights and reports.
6. Support healthcare decision-making.

3.2 Solution Requirements

- Heart Disease Dataset

- Data Cleaning
- Data Preparation
- Tableau Dashboard
- Interactive Charts
- Filters
- Analytical Reports

3.3 Data Flow Diagram

Patient Dataset

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Data Cleaning

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Data Preparation

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Exploratory Data Analysis

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Tableau Dashboard

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Healthcare Insights

3.4 Technology Stack

Programming Language

- Python (Optional for preprocessing)

Libraries

- Pandas
- NumPy

Visualization Tool

- Tableau

Development Tools

- Tableau Desktop / Tableau Public

- Jupyter Notebook
- VS Code
- Microsoft Excel

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution transforms raw patient data into meaningful visual insights using Tableau dashboards. It enables healthcare professionals to analyze disease patterns and make informed decisions efficiently.

4.2 Proposed Solution

Develop an interactive Tableau dashboard that analyzes patient health records and visualizes important factors influencing heart disease, allowing users to explore trends through filters and charts.

4.3 Solution Architecture

Heart Disease Dataset

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Data Cleaning

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Data Preparation

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Exploratory Data Analysis

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Tableau Dashboard

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Insights & Reports

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities
Data Collection	Collect Heart Disease Dataset
Data Preparation	Clean and preprocess data
Data Analysis	Explore trends and patterns
Dashboard Development	Create Tableau dashboard
Testing	Verify dashboard functionality
Documentation	Prepare project documentation

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

- Verified dataset loading.
- Checked data cleaning process.
- Validated calculated fields.
- Tested dashboard filters.
- Verified chart accuracy.
- Confirmed dashboard interactivity.

6.2 Performance Testing

- Tested dashboard responsiveness.
 - Verified filter performance.
 - Ensured correct visualization rendering.
 - Checked report generation speed.
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7. RESULTS

7.1 Dashboard Outputs

Include screenshots of:

- Dashboard Overview
- Heart Disease Distribution

- Age-wise Analysis
 - Gender-wise Analysis
 - Cholesterol Analysis
 - Blood Pressure Analysis
 - Chest Pain Analysis
 - Maximum Heart Rate Analysis
 - Correlation Heatmap
 - Interactive Dashboard with Filters
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8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy understanding of patient health trends.
- Interactive dashboard visualization.
- Faster data analysis.
- Better healthcare decision support.
- User-friendly reports.
- Helps identify disease risk factors quickly.

Disadvantages

- Depends on dataset quality.
 - Limited to available patient records.
 - Requires periodic dataset updates.
 - Cannot replace professional medical diagnosis.
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9. CONCLUSION

The Heart Disease Analysis project successfully demonstrates how data analytics and Tableau visualization can transform healthcare data into meaningful insights. The dashboard enables users to identify important health trends, analyze patient conditions, and support informed healthcare decisions through interactive visualizations.

10. FUTURE SCOPE

- Integration with hospital databases.
 - Real-time healthcare dashboards.
 - AI-assisted healthcare analytics.
 - Cloud-based reporting system.
 - Mobile dashboard application.
 - Predictive healthcare analytics.
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11. APPENDIX

🇮🇹 Tableau Dashboard

Explore the interactive dashboard

🔗 https://public.tableau.com/views/HeartDiseaseAnalysisDashboard_17830875285070/Dashboard3?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

📖 Tableau Story

View the complete Tableau Story here:

🔗 https://public.tableau.com/views/HeartDiseaseAnalysisStory_17830876327310/Story1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

🎥 Demo Video

Watch the project demonstration:

🔗 https://drive.google.com/file/d/1LxF407LHaX7boOF6YmIXhDZZ-TK3mg1z/view?usp=drive_link

📁 Dataset

Download the dataset used for this project:

🔗 https://drive.google.com/file/d/1qoV4wYdp4CAsoH33WbyZWhtixDA_nm61/view?usp=drive_linkhttps://drive.google.com/file/d/1qoV4wYdp4CAsoH33WbyZWhtixDA_nm61/view?usp=drive_link